Yue Gao

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Research Interests

AI Security ML System Security, Adversarial Robustness, Attacks and Defenses, Security and Privacy.
Machine Learning Vision Recognition, LLMs, Multi-Modality, Diffusion Models, Natural Language Processing.
Cybersecurity Multi-component ML Systems, Web-based Applications, Cryptography, Linux Kernel Memory.

Education

University of Wisconsin–Madison — Madison, WI
Ph.D. in Computer Sciences (advised by Prof. Kassem Fawaz) — *Sep 2018 – Mar 2024*
- Thesis: *Challenges and Advances in Adaptive Response Strategies for Machine Learning Security*
- Selected Courses: *Mathematical Foundations of Machine Learning, Advanced Algorithms, Intro to Information Security.*

Shanghai University — Shanghai, China
B.S. in Computer Science and Technology (GPA 3.99/4.00, Ranked 1/292) — *Sep 2014 – Jul 2018*
- Thesis: *A Deep Neural Network-based Image Compression Method*
- Selected Courses: *Operating Systems, Computer Network, Assembly Language, Software Engineering.*

Work Experience

Product Security Engineer @ Snowflake Inc. — San Mateo, CA
Managed by Sharath Sarangpur — *Apr 2024 – Present*
- Red-teaming LLM-based production systems for **prompt injection**, **jailbreaking**, and privacy attacks.
- Developed and deployed **robust and scalable** vulnerability scanners using Python, Docker, and Kubernetes.

Research Assistant @ University of Wisconsin–Madison — Madison, WI
Advised by Prof. Kassem Fawaz — *Nov 2018 – Mar 2024*
- Researched the security of **multi-component ML Systems** deployed in real-world environments.
- Systematized the security analysis of ML-based and web-based systems in **black-box** settings.

ML System Security Research Intern @ Microsoft Research — Redmond, WA
Mentored by Dr. Jay Stokes and Dr. Emre Kiciman — *Jun 2021 – Sep 2021*
- Proposed a research project on defenses against imperceptible textual **backdoor attacks on language models**.
- Discovered blind spots in state-of-the-art attacks and defenses, and published stronger defenses at MILCOM.
- Successfully reduced the attack success rate from 100% to 12%, even at a challenging poisoning rate of 10%.

ML Research and Development Intern @ TuCodec (Startup) — Shanghai, China
Mentored by Dr. Chunlei Cai — *Jan 2018 – Jul 2018*

– *ML Research*
- Secured 1st place as a **primary contributor** in the CVPR 2018 Challenge on Learned Image Compression.
- Improved the average runtime of DNN-based image compression algorithms **from 1 min to 4 secs** per 4K-res image.

– *ML Engineering & Security*
- Independently developed ML-based desktop apps on Ubuntu, MacOS, and Windows using TensorFlow and C++.
- Independently developed ML-based image compression systems using TensorFlow, Python, Docker, and Kubernetes.
- Protected ML systems from **security intrusion** and **model stealing** with security measures and anomaly analysis.

Publications

Conference

[1] **Supply-Chain Attacks in Machine Learning Frameworks**
Yue Gao, Ilia Shumailov, and Kassem Fawaz
The Eighth Annual Conference on Machine Learning and Systems (MLSys), 2025

[2] **SEA: Shareable and Explainable Attribution for Query-based Black-box Attacks**
Yue Gao, Ilia Shumailov, and Kassem Fawaz
Proceedings of the 3rd IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2025

[3] **On the Limitations of Stochastic Pre-processing Defenses**
Yue Gao, Ilia Shumailov, Kassem Fawaz, and Nicolas Papernot
Proceedings of the 36th Conference on Neural Information Processing Systems (NeurIPS), 2022

[4] **Rethinking Image-Scaling Attacks: The Interplay Between Vulnerabilities in Machine Learning Systems**
Yue Gao, Ilia Shumailov, and Kassem Fawaz
Proceedings of the 39th International Conference on Machine Learning (ICML), 2022
Oral Presentation (Top 2%)

[5] **Experimental Security Analysis of the App Model in Business Collaboration Platforms**
Yunang Chen*, *Yue Gao**, Nick Ceccio, Rahul Chatterjee, Kassem Fawaz, and Earlence Fernandes
31st USENIX Security Symposium (USENIX Security), 2022
Bug Bounty ($1500)

[6] **I Know Your Triggers: Defending Against Textual Backdoor Attacks With Benign Backdoor Augmentation**
Yue Gao, Jack W. Stokes, Manoj Prasad, Andrew Marshall, Kassem Fawaz, and Emre Kiciman
IEEE Military Communications Conference (MILCOM), 2022

Workshop

[1] **Variational Autoencoder for Low Bit-rate Image Compression**
Lei Zhou*, Chunlei Cai*, *Yue Gao*, Sanbao Su, and Junmin Wu
Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2018
Winner of the first Challenge on Learned Image Compression

Preprints

[1] **Human-Producible Adversarial Examples**
David Khachaturov, *Yue Gao*, Ilia Shumailov, Robert Mullins, Ross Anderson, and Kassem Fawaz
arXiv, 2023

[2] **Analyzing Accuracy Loss in Randomized Smoothing Defenses**
*Yue Gao**, Harrison Rosenberg*, Kassem Fawaz, Somesh Jha, and Justin Hsu
arXiv, 2020

Selected Projects

Real-world ML Systems Security: Threat Modeling, Defending, and Characterization *Sep 2020 – Jun 2023*
- Investigated the security of **multi-component ML systems** exposed to diverse security threats, e.g., dependencies.
- Revealed **threats amplified by 9x** and **broke state-of-the-art defenses** by jointly exploiting multiple vulnerabilities.
- **Formally proved the non-robustness** of randomization-based defenses beyond demonstrating empirical attacks.
- Characterized the attack's progression for **forensic purposes** and **human-explainable intelligence sharing**.

GARD: Guaranteeing AI Robustness Against Deception *Jun 2019 – Feb 2024*
– *Leadership*
- Led a 9-member cross-university team to **1st and 2nd place in grant competitions hosted by DARPA**.
- Developed initial code bases and mentored **team members with varying technical backgrounds**.
- Onboard and mentored new team members to maintain adversarial mindsets with designing defenses.

– *Individual Contribution*
- Performed **red teaming** and **broke over 10 internal defenses** proposed by team members prior to submission.
- Proposed CLIP-like and diffusion-based methods to **enforce robust features** across RGB and Depth modalities.
- Successfully reduced the disappearance rate from 62% to 9% even **under the red-team evaluation from MITRE**.
- Contributed plug-and-play modules to the official upstream evaluation team and received acknowledgment.

Security Analysis of Online Business Collaboration Platforms *Mar 2021 – Dec 2021*
- Automated the analysis of security principle violations for 3K+ third-party apps in Slack and Microsoft Teams.
- Reverse engineered OAuth designs to bypass access control, and received bug bounty for medium severity.
- Demonstrated POC attacks of eavesdropping on private chats, spoofing video calls, and unauthorized code merging.

Selected Honors & Awards

Slack Bug Bounty: Medium Severity, $1500 — 2022
Top 10% Reviewers Award: NeurIPS — 2022
CVPR Competition Winner: Challenge on Learned Image Compression — 2018
National Scholarship: China — 2017
Top 100 Elite Collegiate Award: China Computer Federation — 2017
Scholarship for Exceptional Leadership: Shanghai University — 2017
City Scholarship: Shanghai — 2016
Outstanding Student Award: Shanghai University — 2016
Outstanding Volunteer Award: ACM ICPC Asia Regional Contest — 2016
Scholarship for Exceptional Innovation: Shanghai University — 2016
Scholarship for Exceptional Academic Achievements: Shanghai University — 2015 – 2018
Bronze Prize for Programming Contest: ACM ICPC Asia East-Continent Final Contest — 2015
Bronze Prize for Programming Contest: ACM ICPC Asia Shanghai Regional Contest — 2015

Professional Activities

Reviewer: NeurIPS, ICML, and ICLR — 2022 – 2024
External Reviewer: IEEE Symposium on Security and Privacy — 2021 – 2023
External Reviewer: USENIX Security Symposium — 2021 – 2022
External Reviewer: ACM Conference on Computer and Communications Security — 2019
Team Leader: Collegiate ICPC Team at Shanghai University — 2016 – 2017

Talks

1. **Forensics and Intelligence Sharing for ML Security** — *Oct 2023*
   DARPA GARD PI Meeting, IBM Research
2. **The Vulnerabilities of Preprocessing in Adversarial Machine Learning** — *Oct 2023*
   ML Red Team, Google
3. **The Vulnerabilities of Preprocessing in Adversarial Machine Learning** — *Apr 2023*
   TrustML Young Scientist Seminar, RIKEN AIP
4. **On the Limitations of Stochastic Pre-processing Defenses** — *Oct 2022*
   DARPA GARD PI Meeting, University of Southern California (virtual)
5. **The Interplay Between Vulnerabilities in Machine Learning Systems** — *Sep 2022*
   DARPA GARD PI Meeting, University of Michigan
6. **Experimental Security Analysis of the App Model in Business Collaboration Platforms** — *Aug 2022*
   USENIX Security 2022
7. **The Interplay Between Vulnerabilities in Machine Learning Systems** — *Jun 2022*
   ICML 2022

Teaching and Mentoring

Project Mentor: DARPA GARD Project, University of Wisconsin–Madison — Fall 2023
Teaching Assistant: CS 368 (C++ for Java Programmers), University of Wisconsin–Madison — Fall 2018
Guest Lecturer: Advanced Algorithms & Data Structures, Shanghai University — 2015 – 2017
Problem Designer: Undergraduate Programming Contests, Shanghai University — 2015 – 2017
Student Mentor: Undergraduate Computer Science Coursework, Shanghai University — 2015 – 2017

Technical Skills

Python Expertise in ML and Security Research (2018 – present) and Backend Development (2016 – 2018).
PyTorch Expertise in ML Research (2019 – present) and Distributed Training (2020 – present).
Docker Expertise in ML Research (2018 – present) with familiarity in Kubernetes Cluster (2017 – present).
C / C++ Proficient in Security Research (2018 – 2019), Software Development (2017 - 2019), and ICPC (2014 – 2018).
Golang Proficient in Software Engineering (2024 – present).
TensorFlow Proficient in ML Research (2017 – 2020) and Service Deployment (2018).

Articles and Media Coverage

CleverHans. Can stochastic pre-processing defenses protect your models? 2022
USENIX login. Experimental Security Analysis of the App Model in Business Collaboration Platforms 2022
Wired. Slack's and Teams' Lax App Security Raises Alarms 2022